

Control of the Endocrine System The endocrine system is one of the master control systems of the body. Like most body systems, the endocrine system is regulated by negative feedback mechanisms that function to maintain homeostasis. Recall that negative feedback, also called feedback inhibition, occurs when an increase in a substance “feeds back” in a way that inhibits the system. Concentrations of hormones, and their effects on other body systems, are controlled in similar ways.

BUILD VOCABULARY

Prefixes The prefixes *anti-* and *ante-* can be easily confused. *Anti-*, as in *antidiuretic*, means “against.” *Ante-*, as in *anterior*, means “before.”

Maintaining Water Balance Water balance in the body is one example of how the endocrine system maintains homeostasis. When you exercise, you sweat and lose water. If this water loss continued, your body would soon become dehydrated. But your hypothalamus contains cells that monitor blood water content. As you lose water, concentrations of dissolved materials in blood rise. The hypothalamus responds in two ways. First, it signals the posterior pituitary gland to release antidiuretic hormone (ADH). ADH molecules are carried to the kidneys, where they slow the removal of water from blood. Second, the hypothalamus generates nerve impulses that produce a sensation of thirst to which we respond if water is available.

When concentrations of dissolved materials fall, as it might after you drink several glasses of water, the pituitary releases less ADH. As ADH concentrations fall, the kidneys remove more water from the blood, restoring its proper concentration. This homeostatic system sets both upper and lower limits for blood water content.

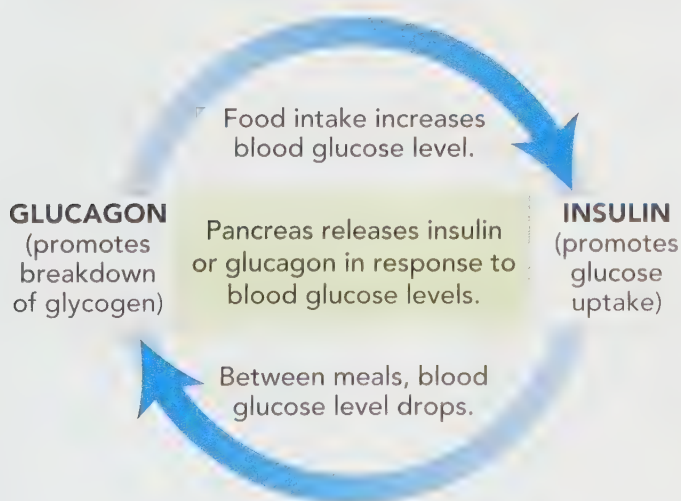
Figure 27-27

Glucagon and Insulin

Insulin and glucagon are opposing hormones that ensure that blood glucose levels stay within a normal range.

Blood Glucose Regulation Glucose concentration in the bloodstream is controlled by insulin and glucagon. When blood glucose concentration rises, the pancreas releases insulin. Insulin stimulates liver and skeletal muscle cells to convert blood glucose to glycogen, and it stimulates fat cells to convert glucose to lipids. These actions together prevent blood glucose concentrations from rising too rapidly. They also store energy for future use. In between meals,

when blood glucose concentration drops, the pancreas releases glucagon. Glucagon stimulates liver and skeletal muscle cells to break down glycogen and release glucose into the blood. These actions raise blood glucose concentrations back to normal. **Figure 27-27** shows this feedback loop.



READING CHECK

Summarize How does the endocrine system respond to rising or falling blood glucose levels?